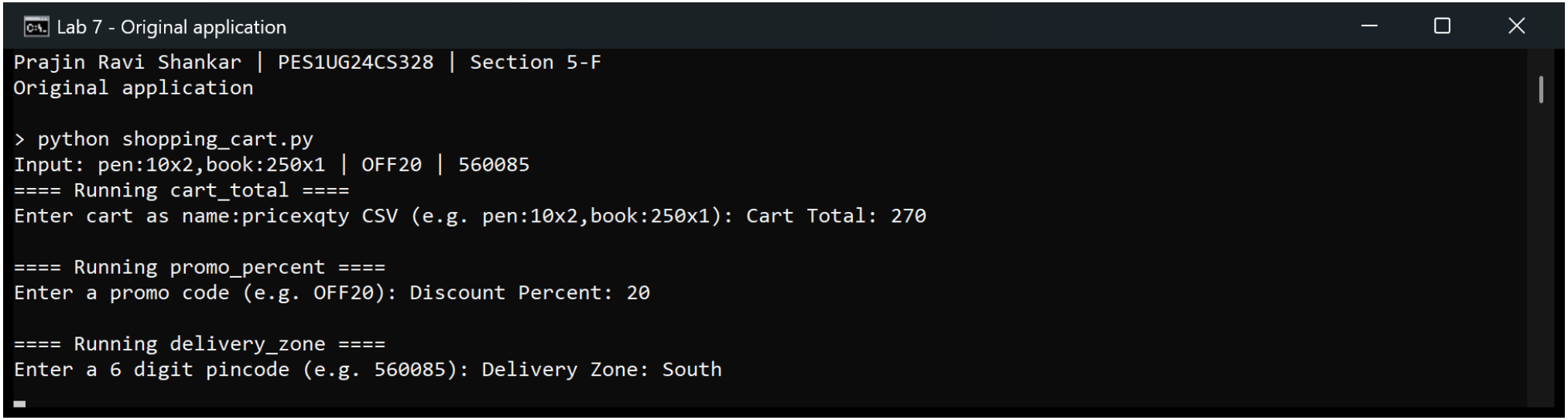
Original application run

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I ran the assigned **shopping_cart.py** with a two-item cart, an OFF20 promo code, and pincode 560085.



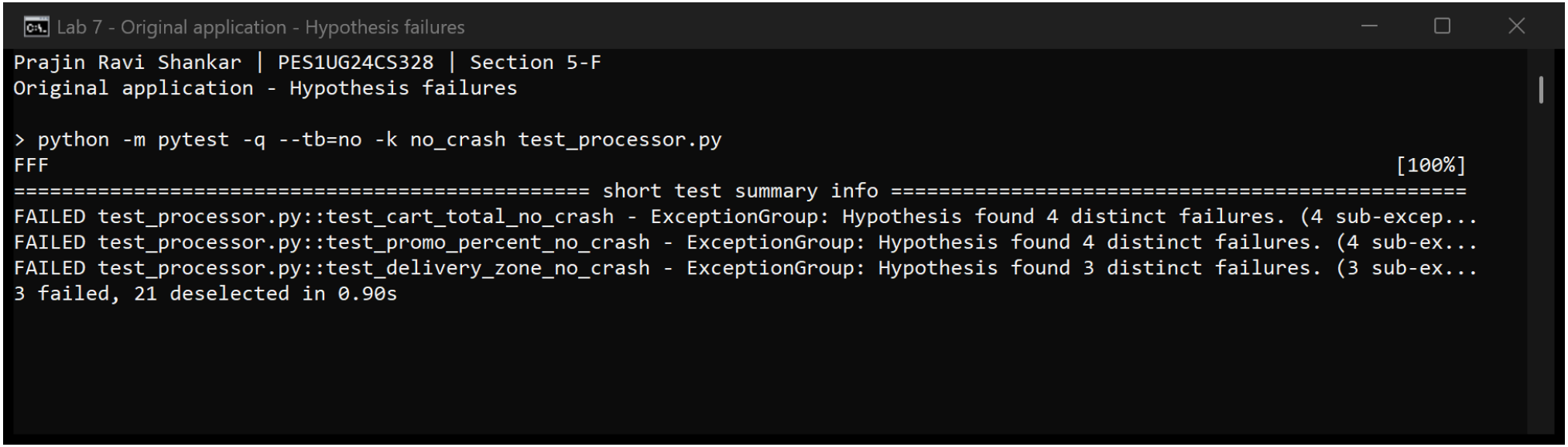
Expected behaviour. The application returned a cart total of 270, a discount of 20%, and the South delivery zone. These outputs provide a reference for checking valid inputs after the fixes.

Assigned source. SETAPESU26/L6_L7_23_shopping_cart, Lab 7. The repository uses **shopping_cart.py** for the application. The updated test script is **test_processor.py** as requested.

Hypothesis checks before the fixes

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I ran the three no-crash properties against the original script. Hypothesis generated text, missing values, integers, and bytes. The terminal shows a failing property for each function.



Cart. An empty cart raised ValueError during splitting. A blank cart now totals 0, and malformed items are rejected. Each valid item needs a name, a non-negative integer price and a positive integer quantity.

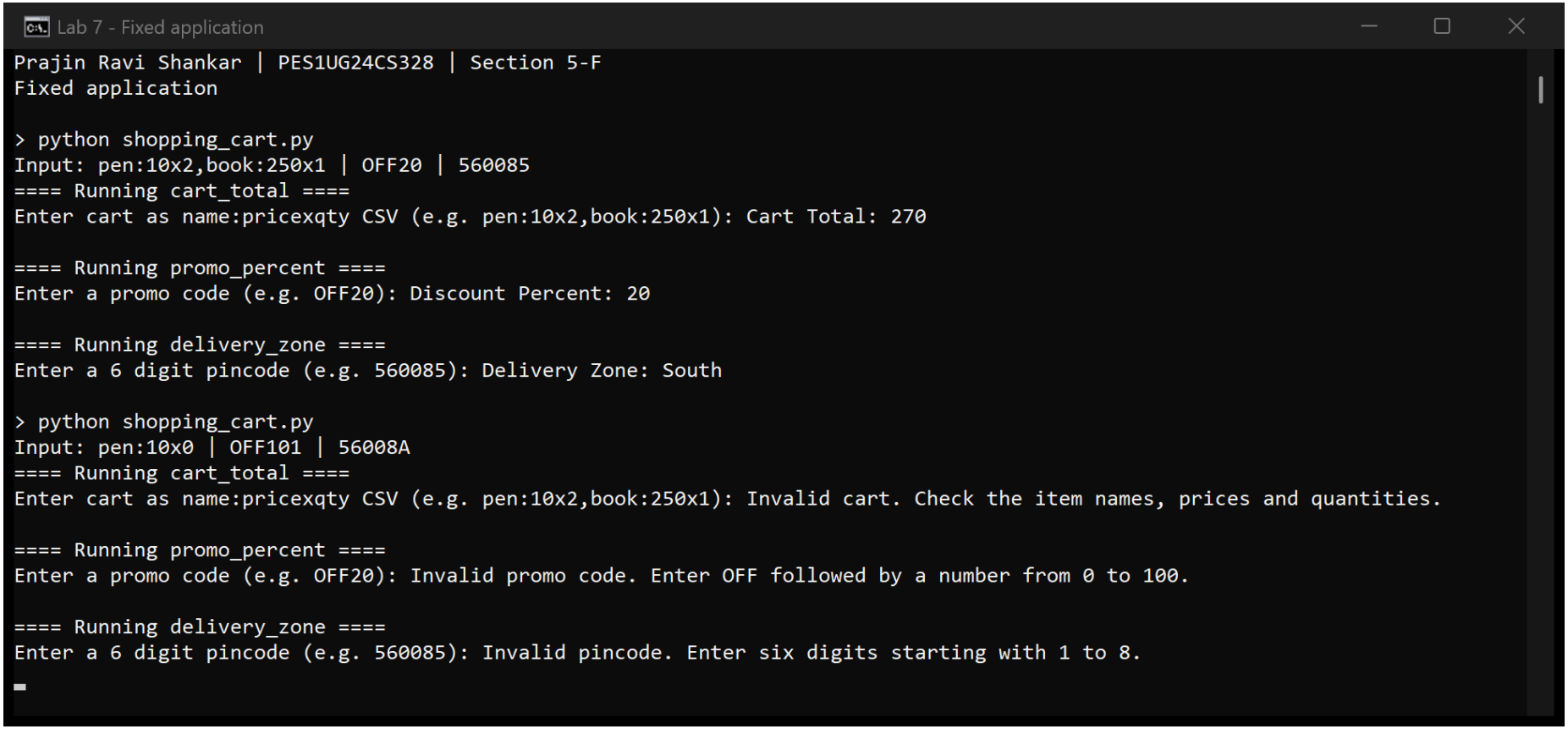
Promo. Blank input caused an IndexError. The fix checks the whole code and accepts case-insensitive OFF followed by a percentage from 0 to 100.

Delivery. Blank input caused an IndexError. The fix requires six ASCII digits and a first digit in the existing zone map (1 through 8).

Fixed application run

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The valid inputs still produce 270, 20%, and South. A second run shows the messages for a zero item quantity, a discount above 100%, and a pincode containing a letter.



Each parser returns **None** for rejected input. The command-line program prints a specific message and continues to the next prompt.

Hypothesis checks after the fixes

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I ran the full updated **test_processor.py** against the fixed application. All 24 pytest cases passed.



Test design. Six Hypothesis properties cover mixed inputs and valid generated carts, promos and pincodes. Each property allows up to 150 generated examples. Eighteen explicit cases check rejection of invalid inputs.

Reproduce. From the **lab7** directory, install the dependencies and run the application and tests:

```
python -m pip install hypothesis pytest
python shopping_cart.py
python -m pytest -q test_processor.py
```